E-trees, two-qubit normal forms, and recursive depth

Draft note

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Abstract

This note introduces a typed formal language of *E-trees* for unitary operators, motivated by the structural observation that a heterogeneous calculus can sometimes be reorganized into a uniform binary-tree syntax. The two-qubit case provides the calibration point: the standard KAK decomposition of SU(4) becomes a canonical representative in the E-tree language on a dense open subset once one fixes a Weyl chamber and local Euler charts. The syntax supports several distinct complexity invariants, including sequential depth, nonlocal node count, Lie depth, and elementary entangling-gate cost. Within an alternating subgrammar, one obtains the structural lower bound $d_{\mathrm{seq}}(t) \geq 2c_{\mathrm{nl}}(t) + 1$. The note then restates the 0/1/2/3-CNOT stratification of two-qubit gates in Cartan coordinates and formulates a recursive higher-qubit program via a minimal recursive E-depth. The main contribution is therefore not a new decomposition theorem, but a representation language and complexity framework that organizes known two-qubit structure and suggests a nontrivial higher-arity research agenda.

0. Introduction

A recent preprint by Odrzywołek observes that the binary operator

$$\mathrm{eml}(x, y) = \exp(x) - \ln(y),$$

together with the constant 1, generates the standard repertoire of elementary functions. In that representation, every elementary expression becomes a binary tree of identical nodes, with grammar

$$S \to 1 \mid \mathrm{eml}(S, S).$$

Whatever one ultimately makes of that observation computationally, it highlights a mathematically clean structural idea: a heterogeneous function calculus can sometimes be reorganized into a uniform recursive syntax tree built from a single binary constructor.

The present note does not study the scalar EML operator itself. Instead, it extracts from that observation a more general question: when a family of transformations admits a recursive binary-tree syntax, can that syntax be used to organize known decomposition theorems and support meaningful complexity invariants? For unitary operators, this question is especially natural, because matrix factorizations already provide a rich supply of local/nonlocal decomposition structures, and because circuit complexity is itself sensitive to how nonlocal structure is assembled.

Our test case is the two-qubit group SU(4). This is the smallest nontrivial setting in which local and entangling structure interact, and it is also the setting in which the Cartan, or KAK, decomposition is completely explicit: every two-qubit unitary is locally equivalent to a unique

canonical interaction

$$A(\alpha,\beta,\gamma) = \exp\big(i(\alpha XX + \beta YY + \gamma ZZ)\big)$$

once one restricts to a Weyl chamber, with nonuniqueness confined to the usual lower-dimensional singular loci. The Bell-basis diagonalization of this Cartan factor makes the geometry especially transparent.

The main contribution of this note is therefore not a new decomposition theorem. Rather, we introduce a formal language of E-trees for unitary operators, show that the standard two-qubit KAK normal form becomes a canonical representative in that language on a dense open subset of SU(4), and define several distinct complexity measures supported by the syntax. In particular, we distinguish

$$d_{\mathrm{seq}} \quad \text{(sequential depth)}, \qquad c_{\mathrm{nl}} \quad \text{(nonlocal node count)}, \qquad d_{\mathrm{Lie}} \quad \text{(commutator depth)},$$

and later also

$$g_{\mathrm{ent}} \quad \text{(elementary entangling-gate cost)}.$$

These invariants live at different levels of description and should not be conflated. One of the advantages of the formal language is that it makes their distinction precise. The resulting framework allows known two-qubit synthesis results to be restated in structural terms, and it also suggests a recursive higher-qubit invariant δ_E whose relation to circuit depth appears to be open.

Two aspects of the note are worth emphasizing at the outset. First, the two-qubit results here are largely a repackaging of standard material: the existence and generic uniqueness statements come from the classical KAK theory for the symmetric pair $(\mathrm{SU}(4), \mathrm{SU}(2)\otimes\mathrm{SU}(2))$, while the 0/1/2/3-CNOT stratification is due to the standard two-qubit synthesis literature. What is new is the formal language in which these results are organized, together with the complexity measures attached to that language. Second, the point of the note is not to claim that a binary-tree grammar automatically yields stronger circuit lower bounds. Rather, it is to isolate a structural invariant that might serve as a complexity witness once one passes beyond the exceptional two-qubit case.

The note is organized as follows. Section 1 introduces the syntax and semantics of E-trees, together with the basic locality and depth notions. Section 2 specializes to two qubits and defines the KAK-shaped normal-form language $\mathcal{N}_2$. Section 3 sketches the proof that this yields a canonical representative on a dense open subset of SU(4). Section 4 introduces an alternating subgrammar and derives a linear lower bound

$$d_{\mathrm{seq}}(t) \geq 2\, c_{\mathrm{nl}}(t) + 1$$

within that restricted syntax. Section 5 then separates grammar-level nonlocality from elementary entangling-gate cost, restates the Weyl-chamber stratification of CNOT complexity in the present language, and formulates the higher-qubit recursive program.

1. Syntax and semantics of E-trees

We introduce a typed formal language for building unitary expressions on n qubits. The purpose of the language is threefold: first, to provide a recursive syntax for local and nonlocal unitary structure; second, to separate syntax from matrix semantics; and third, to support several distinct notions of complexity.

1.1. Qubit arity and primitive symbols

For each $n \geq 1$, let $\mathcal{T}_n$ denote the set of n-qubit E-terms, and let $\mathcal{L}_n$ denote the set of admissible n-qubit Lie expressions.

We begin with primitive skew-Hermitian generators. For one qubit, we include the standard Pauli generators

$$iX, \quad iY, \quad iZ \in \mathcal{L}_1.$$

For two qubits, we include the standard entangling generators

$$iXX, \quad iYY, \quad iZZ \in \mathcal{L}_2.$$

More generally, one may enlarge the primitive set on $\mathcal{L}_n$ to include other k-local skew-Hermitian generators, but for the present note the above set suffices.

1.2. Recursive Lie expressions

We define admissible Lie expressions recursively.

Definition 1.1. *For each n, the set $\mathcal{L}_n$ is the smallest set such that:*

(1) every primitive n-qubit skew-Hermitian generator belongs to $\mathcal{L}_n$;

(2) if $u, v \in \mathcal{L}_n$, then $u + v \in \mathcal{L}_n$;

(3) if $u, v \in \mathcal{L}_n$, then $[u, v] \in \mathcal{L}_n$.

Thus $\mathcal{L}_n$ is not merely a linear span but a recursively generated Lie-expression language. This preserves the nesting structure of commutators and is therefore suitable for defining an algebraic depth notion later.

1.3. Term grammar

We now define the E-terms themselves.

Definition 1.2. *The set $\mathcal{T}_n$ is generated recursively by the following constructors:*

(1) if $u \in \mathcal{L}_n$, then $\exp(u) \in \mathcal{T}_n$;

(2) if $s, t \in \mathcal{T}_n$, then $s \circ t \in \mathcal{T}_n$;

(3) if $s \in \mathcal{T}_m$ and $t \in \mathcal{T}_n$, then $s \otimes t \in \mathcal{T}_{m+n}$.

Here $\circ$ denotes sequential composition and $\otimes$ denotes tensor composition.

This grammar gives a typed syntax in which every term carries a definite qubit arity.

1.4. Matrix semantics

The syntax is interpreted by an evaluation map into the corresponding special unitary group.

Definition 1.3. *For each n, define the semantics map*

$$[\![\cdot]\!] : \mathcal{T}_n \to \mathrm{SU}(2^n)$$

recursively by

$$[\![\exp(u)]\!] = e^{[\![u]\!]}, \qquad [\![s \circ t]\!] = [\![s]\!]\,[\![t]\!], \qquad [\![s \otimes t]\!] = [\![s]\!] \otimes [\![t]\!].$$

For Lie expressions, the interpretation is defined in the evident way: primitive symbols map to their corresponding skew-Hermitian matrices, addition maps to matrix addition, and commutators map to matrix commutators.

This separates formal expressions from the unitaries they represent. Different terms may evaluate to the same matrix.

1.5. Syntactic locality and semantic locality

To state minimality results cleanly, we distinguish a syntactic notion of locality from the corresponding semantic one.

Definition 1.4. *A term $t \in \mathcal{T}_n$ is* syntactically local *if it is generated using only one-qubit primitives, together with the constructors* exp, $\circ$, *and* $\otimes$.

Equivalently, a syntactically local term contains no genuinely multi-qubit primitive generator.

Definition 1.5. *A unitary $U \in \mathrm{SU}(2^n)$ is* semantically local *if U lies in the local subgroup*

$$K_n := \mathrm{SU}(2)^{\otimes n},$$

under the standard embedding.

Lemma 1.6. *If $t \in \mathcal{T}_n$ is syntactically local, then $[\![t]\!]$ is semantically local.*

Proof. Every primitive occurring in t acts on a single qubit. Exponentiating such primitives yields one-qubit special unitaries. Sequential compositions of one-qubit terms remain one-qubit unitaries on the corresponding tensor factors, and tensor products of local unitaries remain local. By induction on the construction of t, the represented unitary lies in K_n. $\square$

For the two-qubit case this immediately implies that any term representing a generic element of SU(4) must contain at least one syntactically nonlocal primitive.

1.6. Three complexity measures

The recursive syntax supports several inequivalent notions of complexity. We record three that will be useful below.

Definition 1.7. *The* sequential depth $d_{\mathrm{seq}} : \mathcal{T}_n \to \mathbb{N}$ *is defined recursively by*

$$d_{\mathrm{seq}}(\exp(u)) = 1,$$

$$d_{\mathrm{seq}}(s \circ t) = 1 + \max\{d_{\mathrm{seq}}(s), d_{\mathrm{seq}}(t)\},$$

$$d_{\mathrm{seq}}(s \otimes t) = \max\{d_{\mathrm{seq}}(s), d_{\mathrm{seq}}(t)\}.$$

This measures serial composition depth.

Definition 1.8. *The* nonlocal count $c_{\mathrm{nl}} : \mathcal{T}_n \to \mathbb{N}$ *counts the number of exponential nodes* $\exp(u)$ *whose Lie expression* u *contains at least one genuinely multi-qubit primitive. Equivalently,* c_{nl} *counts the explicitly nonlocal exponential layers in the tree.*

Definition 1.9. *The* Lie depth $d_{\mathrm{Lie}} : \mathcal{L}_n \to \mathbb{N}$ *is defined recursively by*

$$d_{\mathrm{Lie}}(p) = 0$$

for primitive p*,*

$$d_{\mathrm{Lie}}(u + v) = \max\{d_{\mathrm{Lie}}(u), d_{\mathrm{Lie}}(v)\},$$

$$d_{\mathrm{Lie}}([u, v]) = 1 + \max\{d_{\mathrm{Lie}}(u), d_{\mathrm{Lie}}(v)\}.$$

This is an algebraic nesting depth for commutator structure.

Remark 1.10. The three measures above should not be conflated. In particular, Lie depth records nested commutator structure inside exponential nodes, while nonlocal count records the number of explicitly entangling exponential layers. The latter is the quantity most directly comparable to gate-based notions of entangling complexity; the former is closer to BCH-type algebraic complexity.

2. Two-qubit normal forms

We now specialize to two qubits and define a normal-form sublanguage corresponding to the usual KAK decomposition, expressed in E-tree syntax.

2.1. The local one-qubit chart

Fix a one-qubit Euler chart on a dense open subset of SU(2). For definiteness, let

$$E(a,b,c) = e^{iaZ}e^{ibY}e^{icZ},$$

with

$$a, c \in (-\pi, \pi), \qquad b \in (0, \pi).$$

This excludes the familiar Euler-angle singular set and yields a one-to-one chart on a dense open subset of SU(2). We regard $E(a,b,c)$ as a distinguished family of terms in $\mathcal{T}_1$, and write

$$\mathcal{E}_1 := \{E(a,b,c)\}.$$

2.2. The local two-qubit fragment

Definition 2.1. *The* local chart fragment $\mathcal{K}_2 \subset \mathcal{T}_2$ *is*

$$\mathcal{K}_2 := \{u \otimes v : u, v \in \mathcal{E}_1\}.$$

Each element of $\mathcal{K}_2$ *depends on six real parameters and evaluates into the local subgroup*

$$K_2 = \mathrm{SU}(2) \otimes \mathrm{SU}(2) \subset \mathrm{SU}(4).$$

2.3. The Cartan interaction fragment

Let

$$\mathfrak{a} := \mathrm{span}_{\mathbb{R}}\{iXX, iYY, iZZ\} \subset \mathfrak{su}(4).$$

Since XX, YY, and ZZ mutually commute, $\mathfrak{a}$ is an abelian subalgebra. The corresponding interaction unitaries are therefore of the form

$$A(\alpha,\beta,\gamma) = \exp\big(i(\alpha XX + \beta YY + \gamma ZZ)\big).$$

Fix a Weyl chamber

$$C = \{(\alpha,\beta,\gamma) : \tfrac{\pi}{4} \geq \alpha \geq \beta \geq |\gamma| \geq 0\}.$$

Definition 2.2. *The* Cartan fragment $\mathcal{A}_2 \subset \mathcal{T}_2$ *is*

$$\mathcal{A}_2 := \left\{\exp\big(i(\alpha XX + \beta YY + \gamma ZZ)\big) : (\alpha,\beta,\gamma) \in C\right\}.$$

The interior C° *of the Weyl chamber corresponds to the generic nondegenerate case.*

2.4. Bell-basis diagonalization

The Cartan fragment is especially transparent in the Bell basis. Let Q denote the change-of-basis matrix from the computational basis to the Bell basis. Then

$$Q^\dagger A(\alpha,\beta,\gamma)Q = \mathrm{diag}(e^{i\lambda_1}, e^{i\lambda_2}, e^{i\lambda_3}, e^{i\lambda_4}),$$

where

$$\lambda_1 = \alpha - \beta + \gamma, \qquad \lambda_2 = -\alpha + \beta + \gamma,$$
$$\lambda_3 = \alpha + \beta - \gamma, \qquad \lambda_4 = -\alpha - \beta - \gamma.$$

These phases satisfy

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 0,$$

as required by the determinant-one constraint.

The Weyl walls are precisely the loci where two of the λ_i coincide. Thus the chamber boundary is the singular locus for generic uniqueness of Cartan parameters.

2.5. The two-qubit normal-form language

Definition 2.3. *The* two-qubit normal-form language $\mathcal{N}_2 \subset \mathcal{T}_2$ *is*

$$\mathcal{N}_2 := \{k_L \circ a \circ k_R : k_L, k_R \in \mathcal{K}_2,\ a \in \mathcal{A}_2\}.$$

Thus every term in $\mathcal{N}_2$ has the form

$$(E(a_1,b_1,c_1) \otimes E(a_2,b_2,c_2)) \circ A(\alpha,\beta,\gamma) \circ (E(a_3,b_3,c_3) \otimes E(a_4,b_4,c_4)).$$

This is a 15-parameter family:

$$6 \text{ local-left} + 3 \text{ Cartan} + 6 \text{ local-right} = 15,$$

matching

$$\dim \mathrm{SU}(4) = 15.$$

2.6. Generic normal-form theorem

Proposition 2.4. *There exists a dense open subset* $\mathrm{SU}(4)^\circ \subset \mathrm{SU}(4)$ *such that every* $U \in \mathrm{SU}(4)^\circ$ *is represented by a unique term in* $\mathcal{N}_2$.

More explicitly, for every $U \in \mathrm{SU}(4)^\circ$ *there exist unique local factors* $k_L, k_R \in \mathcal{K}_2$ *and a unique Cartan factor* $A(\alpha,\beta,\gamma) \in \mathcal{A}_2$ *with* $(\alpha,\beta,\gamma) \in C^\circ$ *such that*

$$U = [\![k_L \circ A(\alpha,\beta,\gamma) \circ k_R]\!].$$

The excluded set consists of:

(1) points whose Cartan coordinates lie on the Weyl walls;

(2) points whose local factors fall on the singular locus of the chosen Euler chart.

Remark 2.5. This proposition is not a new decomposition theorem. Its content is rather that, after fixing a Weyl chamber and fixed local Euler charts, the usual two-qubit KAK decomposition induces a canonical representative in the formal language $\mathcal{N}_2$ on a dense open set.

3. Proof sketch of Proposition 2.4

The proof is the standard KAK argument for the symmetric pair

$$(\mathrm{SU}(4),\, \mathrm{SU}(2) \otimes \mathrm{SU}(2)),$$

translated into the language of E-terms. We sketch the main steps and indicate where the genericity assumptions enter.

3.1. Bell-basis conjugation

Let Q denote the change-of-basis matrix from the computational basis to the Bell basis, and set

$$U_B := Q^\dagger U Q \in \mathrm{SU}(4).$$

The Bell basis is distinguished because the commuting generators XX, YY, and ZZ are simultaneously diagonal there.

3.2. Extraction of the Cartan coordinates

Form the matrix

$$M := U_B^T U_B.$$

This is a symmetric unitary matrix. In the standard KAK analysis, the eigenvalues of M are of the form

$$e^{2i\lambda_1},\ e^{2i\lambda_2},\ e^{2i\lambda_3},\ e^{2i\lambda_4},$$

where the λ_j are the Bell-basis phases of the Cartan factor $A(\alpha, \beta, \gamma)$. By the formulas of Section 2, once the eigenphases are known, the map from $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ to (α, β, γ) is linear. Restricting to the interior C° of the chosen Weyl chamber makes this correspondence unique.

Accordingly, for U outside the Weyl-wall locus, there is a unique triple $(\alpha, \beta, \gamma) \in C^\circ$ whose associated Cartan factor has the same Bell-basis phase data as M.

3.3. Reconstruction of the local factors

Having fixed the Cartan factor $A = A(\alpha, \beta, \gamma)$, the KAK theorem (see Helgason, Chapter IX) implies the existence of local factors

$$k_L,\ k_R \in K_2 := \mathrm{SU}(2) \otimes \mathrm{SU}(2)$$

such that

$$U = k_L A k_R.$$

This reconstruction is genuinely two-sided: in general neither UA^{-1} nor $A^{-1}U$ is local. The local factors must therefore be solved for simultaneously, rather than extracted by one-sided division.

On the dense open subset where the Cartan coordinates lie in C°, the residual ambiguity in (k_L, k_R) is given by the stabilizer of A inside K_2. For generic Cartan triples this stabilizer is finite; after fixing the Euler chart on each SU(2) factor, the ambiguity disappears away from the chart singular loci.

3.4. Passage to the E-tree normal form

Each local factor in K_2 can be written as a tensor product

$$k_L = u_1 \otimes u_2, \qquad k_R = u_3 \otimes u_4,$$

with $u_j \in \mathrm{SU}(2)$. On the dense open subset where each u_j avoids the singular locus of the chosen Euler chart, there exist unique parameters (a_j, b_j, c_j) such that

$$u_j = E(a_j, b_j, c_j).$$

Substituting these into the decomposition

$$U = (u_1 \otimes u_2)\, A(\alpha, \beta, \gamma)\, (u_3 \otimes u_4)$$

produces a unique representing term in $\mathcal{N}_2$. This proves the proposition.

3.5. Exceptional loci

The excluded subset of SU(4) is the union of two lower-dimensional loci:

(1) the Weyl-wall locus, where the Bell-basis phases collide and the Cartan coordinates cease to be uniquely determined by the chosen chamber;

(2) the Euler-chart singular locus for one or more of the four local SU(2) factors.

Their complement is dense and open, yielding the claimed generic uniqueness.

Remark 3.1. The proof exhibits an intrinsic local–nonlocal–local structure:

$$K_2 \cdot A \cdot K_2.$$

This alternation is not built into the ambient grammar $\mathcal{T}_2$, but it emerges canonically from the decomposition theorem. It therefore provides a natural motivation for imposing an alternation constraint when passing to restricted E-tree grammars.

4. Restricted grammars and alternation constraints

The normal-form language $\mathcal{N}_2$ has an intrinsic structural property visible in the proof above: every term in $\mathcal{N}_2$ consists of a local layer, followed by a single nonlocal Cartan node, followed by another local layer. No two nonlocal nodes appear consecutively. We call this the alternation property.

This section asks three questions. First, can we define a restricted subgrammar of $\mathcal{T}_2$ that enforces alternation syntactically? Second, is this subgrammar still generically universal for SU(4)? Third, does alternation sharpen the minimality and uniqueness claims of the previous section?

4.1. The alternating subgrammar

Definition 4.1. *A term $t \in \mathcal{T}_n$ is* alternating *if, in every maximal sequential chain of its construction tree, nonlocal exponential nodes and syntactically local blocks strictly alternate, beginning and ending with local blocks.*

More precisely, define the alternating fragment $\mathrm{Alt}_n \subseteq \mathcal{T}_n$ inductively as follows:

(i) every syntactically local term is in Alt_n;

(ii) if k_L, k_R are syntactically local two-qubit terms and $a \in \mathcal{A}_2$, then $k_L \circ a \circ k_R \in \mathrm{Alt}_2$;

(iii) for $n > 2$, if $t_L \in \mathrm{Alt}_j$ and $t_R \in \mathrm{Alt}_k$ with $j + k = n$, then $t_L \otimes t_R \in \mathrm{Alt}_n$;

(iv) if $s, t \in \mathrm{Alt}_n$, then $s \circ t \in \mathrm{Alt}_n$ provided the resulting composition still satisfies the alternation condition and does not place two nonlocal nodes consecutively at the junction.

Remark 4.2. The normal-form language $\mathcal{N}_2$ is contained in Alt_2 by construction. The inclusion is strict: Alt_2 also contains sequential compositions of $\mathcal{N}_2$-terms, provided the alternation condition is preserved.

4.2. Generic universality of the alternating subgrammar

Proposition 4.3. *The alternating subgrammar Alt_2 is generically universal for SU(4): there exists a dense open subset $U \subseteq \mathrm{SU}(4)$ such that every element of U is represented by a term in Alt_2.*

Proof. By Section 2, every element of the dense open set constructed there is represented by a term in $\mathcal{N}_2$. Since $\mathcal{N}_2 \subseteq \mathrm{Alt}_2$, the same dense open set witnesses generic universality of Alt_2. □

Remark 4.4. Imposing alternation therefore does not reduce expressiveness at the one-Cartan-node level. The more interesting question is whether alternation sharpens structural statements once one allows representations with larger nonlocal count.

4.3. Alternation and structural lower bounds

Lemma 4.5. *In the alternating subgrammar* Alt_2*, any term whose maximal sequential chain contains* k *nonlocal nodes must contain at least* $k+1$ *local blocks separating them and anchoring the two ends.*

Proof. By the alternation condition, no two nonlocal nodes may occur consecutively in a sequential chain. Hence between any two adjacent nonlocal nodes there must be at least one local block, and there must also be a local block at each end of the chain. Therefore a chain with k nonlocal nodes requires at least $k-1$ internal local separators plus 2 endpoint local blocks, for a total of $k+1$ local blocks. □

Corollary 4.6. *If* $t \in \mathrm{Alt}_2$*, then*

$$d_{\mathrm{seq}}(t) \geq 2\, c_{\mathrm{nl}}(t) + 1.$$

For the normal form $\mathcal{N}_2$*, one has* $c_{\mathrm{nl}} = 1$ *and* $d_{\mathrm{seq}} = 3$*, so the bound is saturated.*

Proof. Let $c_{\mathrm{nl}}(t) = k$. By Lemma 4.5, any maximal sequential chain of t containing all k nonlocal nodes must contain at least $k+1$ local blocks separating them and anchoring the two ends. Each such local block contributes at least 1 to d_{seq}, and each nonlocal node also contributes at least 1. Hence

$$d_{\mathrm{seq}}(t) \geq k + (k+1) = 2k + 1 = 2c_{\mathrm{nl}}(t) + 1.$$

□

Remark 4.7. The bound above is a structural statement about alternating grammars: sequential depth grows at least linearly with nonlocal node count. It should not be interpreted as a statement about elementary gate count. In particular, a single nonlocal node may itself require several elementary entangling gates when compiled into a discrete circuit model.

4.4. Alternating normal form

Definition 4.8. *A term* $t \in \mathrm{Alt}_n$ *is in* alternating normal form *if:*

(1) t *contains the minimum number of nonlocal nodes* $c_{\mathrm{nl}}(t)$ *among all alternating terms representing* $[\![t]\!]$*;*

(2) each local layer is expressed in the fixed one-qubit Euler chart $\mathcal{E}_1$*;*

(3) each Cartan-type nonlocal node is expressed using coordinates in the chosen chamber interior whenever such a chamber description is available.

Remark 4.9. For two qubits, the alternating normal form with $c_{\mathrm{nl}} = 1$ is exactly $\mathcal{N}_2$, and by the global KAK decomposition this already suffices to represent every element of SU(4). Thus, for two qubits, higher nonlocal counts are not required for expressiveness, but only arise in non-normal-form representations or in finer-grained decompositions into elementary entangling gates. The case $c_{\mathrm{nl}} \geq 2$ becomes structurally essential only in larger systems, where no single two-qubit Cartan node can capture the full nonlocal content.

5. Entangling-gate cost and recursive extension

5.1. Elementary entangling-gate cost

The nonlocal count c_{nl} introduced above is a grammar-level invariant: it counts the number of syntactically nonlocal exponential nodes in an E-tree. It does not directly measure the cost of implementing those nodes in a physical gate set. We now introduce a complementary invariant that captures synthesis cost.

Definition 5.1. *Let $U \in \mathrm{SU}(2^n)$. The* elementary entangling-gate cost $g_{\mathrm{ent}}(U)$ *is the minimum number of two-qubit elementary gates (here, CNOT gates) required to implement U, optimized over all circuits that implement U up to local one-qubit unitaries on each qubit.*

The phrase "up to local one-qubit unitaries" reflects the KAK structure: local layers contribute zero entangling cost, and the Cartan node carries all nonlocal content. This makes g_{ent} naturally a function on local-equivalence classes rather than on raw matrix factorizations.

Remark 5.2. The two invariants c_{nl} and g_{ent} are related but operate at different levels of description. The former counts entangling layers in the E-tree grammar, while the latter counts elementary two-qubit gates in a physical implementation relative to a chosen gate set. For two-qubit unitaries in KAK normal form, $c_{\mathrm{nl}} = 1$ throughout, while $g_{\mathrm{ent}} \in \{0, 1, 2, 3\}$ depending on the Cartan coordinates.

5.2. Two-qubit gate-cost stratification inside one Cartan node

For two qubits, every unitary $U \in \mathrm{SU}(4)$ is locally equivalent to a unique Cartan representative

$$A(\alpha, \beta, \gamma) = \exp\big(i(\alpha XX + \beta YY + \gamma ZZ)\big),$$

with

$$\frac{\pi}{4} \geq \alpha \geq \beta \geq |\gamma| \geq 0,$$

away from the Weyl-wall singular set. Thus the elementary entangling-gate cost of a two-qubit unitary can be read off from the Cartan coordinates of its single nonlocal node, not from the nonlocal node count c_{nl}, which stays equal to 1 throughout the entire nonlocal KAK family.

Theorem 5.3. *Let $U \in \mathrm{SU}(4)$, and let (α, β, γ) be its Cartan coordinates in the chosen Weyl chamber. Let $g_{\mathrm{ent}}(U)$ denote the minimum number of CNOT gates needed to implement U up to local one-qubit unitaries. Then:*

(1) ***Zero CNOTs:***

$$g_{\mathrm{ent}}(U) = 0 \quad \Longleftrightarrow \quad \alpha = \beta = \gamma = 0.$$

(2) ***One CNOT:***

$$g_{\mathrm{ent}}(U) = 1 \quad \Longleftrightarrow \quad \beta = \gamma = 0,\ \alpha \neq 0.$$

Equivalently, U is locally equivalent to a controlled-phase, hence controlled-V, gate.

(3) ***Two CNOTs:***

$$g_{\mathrm{ent}}(U) = 2 \quad \Longleftrightarrow \quad \gamma = 0,\ \beta > 0.$$

(4) ***Three CNOTs:***

$$g_{\mathrm{ent}}(U) = 3 \quad \Longleftrightarrow \quad \gamma > 0.$$

In particular, the region $\gamma > 0$ requires exactly three CNOT gates, while the two-CNOT locus is the codimension-one face $\gamma = 0$, and the one-CNOT locus is the edge $\beta = \gamma = 0$.

Remark 5.4 (Example: the controlled-Z gate). The controlled-Z gate has Cartan coordinates

$$(\alpha, \beta, \gamma) = \left(\frac{\pi}{4}, 0, 0\right),$$

so it lies on the edge $\beta = \gamma = 0$ with $\alpha = \pi/4 \neq 0$. By case (2) of the theorem,

$$g_{\mathrm{ent}}(CZ) = 1.$$

Proof sketch. The case $g_{\mathrm{ent}} = 0$ is exactly the local subgroup $K_2 = \mathrm{SU}(2) \otimes \mathrm{SU}(2)$, corresponding to the identity Cartan point. Vidal and Dawson show that a two-qubit gate can be implemented with two CNOT gates if and only if, in canonical coordinates, the third Cartan parameter vanishes. In the ordered Weyl chamber used here, this is precisely the condition $\gamma = 0$. They also identify the more special controlled-unitary edge, corresponding to one nonzero Cartan parameter only. Finally, Vatan–Williams and Shende–Markov–Bullock show that three CNOT gates are necessary and sufficient for an arbitrary two-qubit gate, so outside the two-CNOT locus one is in the generic three-CNOT region. □

Remark 5.5. This theorem makes precise the distinction between grammar-level and gate-level complexity. For every nonlocal two-qubit unitary in KAK normal form, the syntactic invariant satisfies

$$c_{\mathrm{nl}} = 1,$$

because there is exactly one Cartan node. Yet the elementary entangling-gate cost g_{ent} takes the values 1, 2, or 3 depending on where that single Cartan node lies inside the Weyl chamber. Thus c_{nl} records the number of nonlocal layers in the representation language, while g_{ent} records the discrete synthesis cost of compiling a given nonlocal layer into a chosen elementary gate set.

Remark 5.6. Geometrically, the CNOT-cost stratification is nested inside the Weyl chamber:

- the origin is the local class;
- the edge $(\alpha, 0, 0)$ is the one-CNOT family;
- the face $(\alpha, \beta, 0)$ is the two-CNOT family;
- the region (α, β, γ) with $\gamma > 0$ is the three-CNOT region.

5.3. Recursive extension to n qubits

The two-qubit case is exceptional in that all nonlocal structure can be concentrated into a single Cartan node. For larger systems, no single two-qubit interaction layer can capture the full nonlocal content of a generic unitary. This suggests that the correct higher-arity generalization is recursive rather than flat.

The guiding idea is to build n-qubit unitaries by repeatedly decomposing them into two lower-arity blocks, together with a nonlocal interaction layer coupling those blocks. In this way, the formal language records not only the presence of nonlocality, but also the recursive architecture by which it is assembled.

Definition 5.7. *Let $t \in \mathrm{Alt}_n$. Define the* recursive E-depth $\delta_E(t)$ *by the following rules:*

(1) ***Local base case.*** *If t is syntactically local, then*

$$\delta_E(t) = 0.$$

(2) ***Tensor composition.*** *If $t = s \otimes r$, then*

$$\delta_E(t) = \max\{\delta_E(s), \delta_E(r)\}.$$

(3) ***Local sequential composition.*** *If $t = s \circ r$ with $s, r \in \mathrm{Alt}_n$ and $t \in \mathrm{Alt}_n$, and the composition introduces no new nonlocal interaction layer, then*

$$\delta_E(t) = \max\{\delta_E(s), \delta_E(r)\}.$$

(4) ***Nonlocal coupling step.*** *If t is obtained by coupling two lower-arity alternating terms $s \in \mathrm{Alt}_j$ and $r \in \mathrm{Alt}_{n-j}$ by a genuinely nonlocal interaction layer, then*

$$\delta_E(t) = 1 + \max\{\delta_E(s), \delta_E(r)\}.$$

The last clause is schematic: its precise realization depends on the chosen higher-arity interaction grammar. Its purpose is to record the number of recursive levels at which nonlocal couplings are introduced.

Remark 5.8. The invariant δ_E is neither the same as the nonlocal count c_{nl} nor the same as the elementary entangling-gate cost g_{ent}. Rather, c_{nl} counts nonlocal nodes in a fixed representation, g_{ent} counts elementary entangling gates in a discrete synthesis model, and δ_E measures the recursive depth at which nonlocal structure is assembled.

Definition 5.9. *For $U \in \mathrm{SU}(2^n)$, define the* minimal recursive E-depth

$$\delta_E(U) := \min\{\delta_E(t) : t \in \mathrm{Alt}_n,\ [\![t]\!] = U\}.$$

Thus $\delta_E(U)$ is a structural invariant of the unitary relative to the chosen alternating grammar.

Remark 5.10. For $n = 2$, the KAK normal form trivializes the recursive picture: every unitary is representable with a single nonlocal Cartan node, so the higher-level recursive structure collapses. The invariant δ_E therefore becomes genuinely interesting only for $n > 2$.

Conjecture 5.11. *For a suitable recursive alternating grammar on n-qubit unitaries, the minimal recursive E-depth $\delta_E(U)$ should be a lower bound, up to model-dependent constants, on entangling circuit depth.*

More precisely, one expects constants $a, b > 0$, depending on the chosen elementary gate set and compilation model, such that

$$a\,\delta_E(U) - b \;\leq\; \mathrm{depth}_{\mathrm{ent}}(U)$$

for broad classes of n-qubit unitaries U, where $\mathrm{depth}_{\mathrm{ent}}(U)$ denotes the minimum depth of a circuit realizing U using elementary entangling gates and arbitrary local gates.

Remark 5.12. The conjectural relationship above is plausible because each increment of δ_E records a new level at which previously separate subsystems become nonlocally coupled. In a circuit model, such couplings cannot in general be generated without additional entangling depth. The difficulty is that recursive algebraic structure and discrete gate depth live in different categories: one is a property of a formal factorization language, the other of a compiled circuit over a fixed gate set. Establishing a quantitative bridge between them is therefore nontrivial.

Problem 5.13. *Formulate a higher-arity interaction grammar that satisfies the following three conditions:*

(1) it is expressive enough to represent a dense open subset of $\mathrm{SU}(2^n)$;

(2) it admits alternating normal forms with manageable nonuniqueness;

(3) it yields a recursive depth invariant δ_E that is nontrivial and plausibly comparable to entangling circuit depth.

Problem 5.14. *Determine whether $\delta_E(U)$ gives stronger lower bounds for structured unitary families, such as those generated by k-local Hamiltonians, low-treewidth interaction graphs, or recursively compositional ansätze, than are obtainable from coarse parameter counting alone.*

Remark 5.15. The point of the recursive program is not merely to re-express known synthesis results. Rather, it is to ask whether the architecture of a unitary factorization can itself serve as

a complexity witness. In the two-qubit case, KAK collapses all nonlocality into a single layer; in larger systems, no such collapse is available generically, and recursive E-depth may therefore capture genuinely new structure.

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